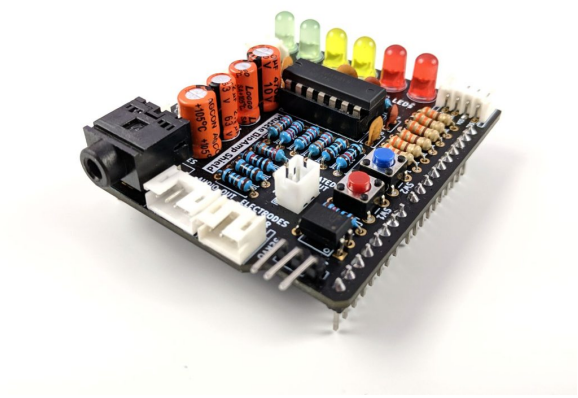
Muscle BioAmp Shield

All-in-one ElectroMyography (EMG) shield for Arduino UNO

Overview

Muscle BioAmp Shield is an all-in-one Arduino Uno ElectroMyography (EMG) shield for learning neuroscience with ease which is inspired from BackYard Brains (BYB) Muscle Spiker shield and provides similar features like hobby servo output, user buttons, LED Bar, Audio output, and battery input. It is perfect for beginners as they can easily stack it on top of Arduino Uno to record, visualize and listen to their muscle signals to make amazing projects in the domain of Human-Computer Interface (HCI).



Features

- Stacks directly on top of Arduino Uno
- Hobby servo output, user buttons, LED bar, audio output and battery input
- STEMMA I2C interface to connect hundreds of extension boards like OLED screens, character displays, accelerometers and servo controllers
- STEMMA analog port to connect an additional BioAmp EXG Pill or any other sensor with analog output
- STEMMA digital port to connect any digital sensor or actuator of your choice

Specifications

Input Voltage	5 V
Input Impedance	10^{11} ohm
Fixed Gain	x2420
Bandpass Filter	72 – 720 Hz
Compatible Hardware	Arduino UNO
BioPotentials	EMG (Electromyography)
No. of Channels	1
Electrodes	3 (Positive, Negative, and Reference)
Dimensions	6.0 x 5.3 cm
Open Source	Hardware + Software

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1 Muscle BioAmp Shield

v0.3

1.4 Hardware

Images below shows a quick overview of the hardware design.

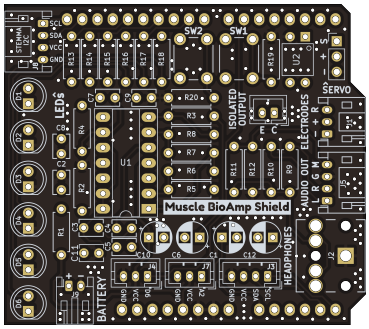


Fig. 1: *PCB Front*

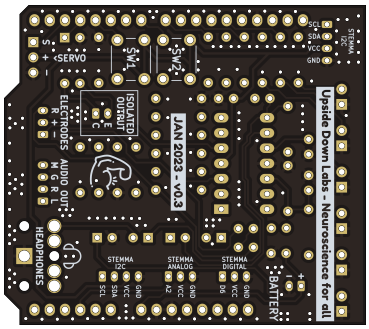


Fig. 2: *PCB Back*

1.5 Contents of the kit

There are 2 variants available for Muscle BioAmp Shield v0.3 kit - one comes with the shield assembled and the other one contains bare PCB of the sensor and the components separately which you can assemble pretty easily.

1.6 Software requirements

1. **Arduino IDE**
   - Latest Version (v2.3.8+): Arduino IDE Latest Version Download *(Recommended for faster compilation and enhanced support for the latest Arduino UNO Development Boards)*
   - Legacy Version (v1.8.19): Arduino IDE v1.8.19 (legacy IDE) Download *(For Real-time signal visualization using a better built-in Serial Plotter)*
2. **Chords Web Visualizer**

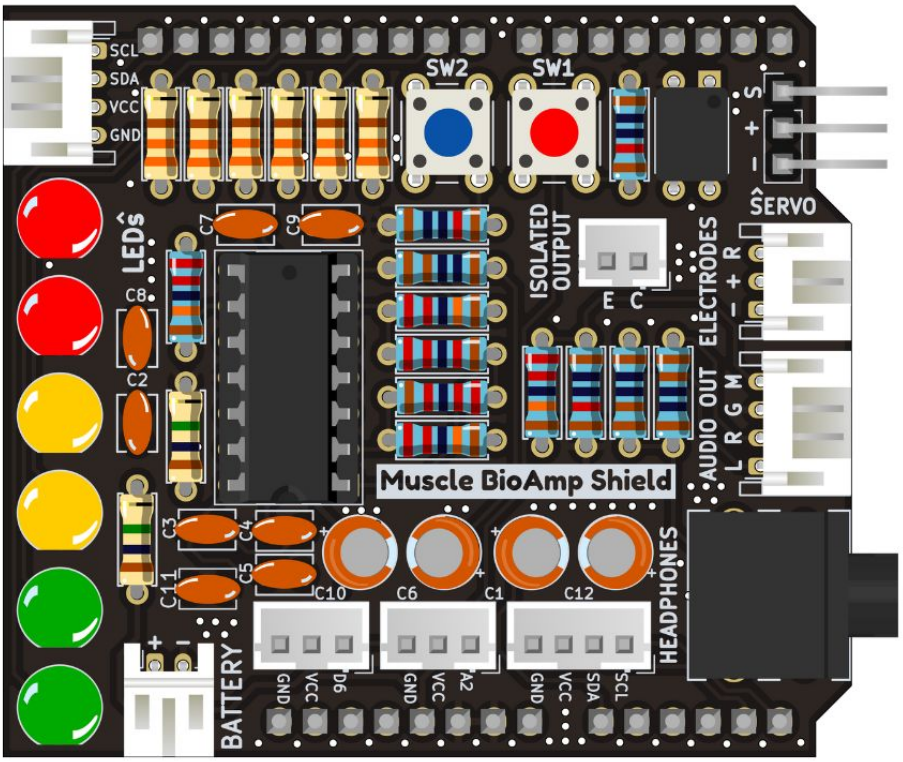


Fig. 3: *Assembled PCB*

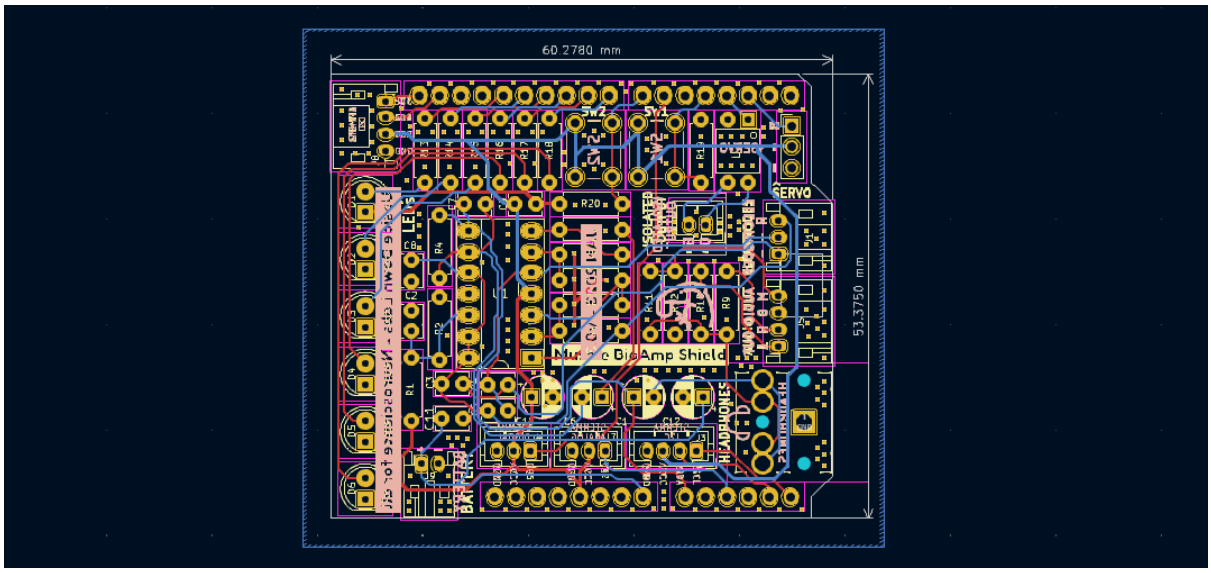


Fig. 4: *PCB Layout*

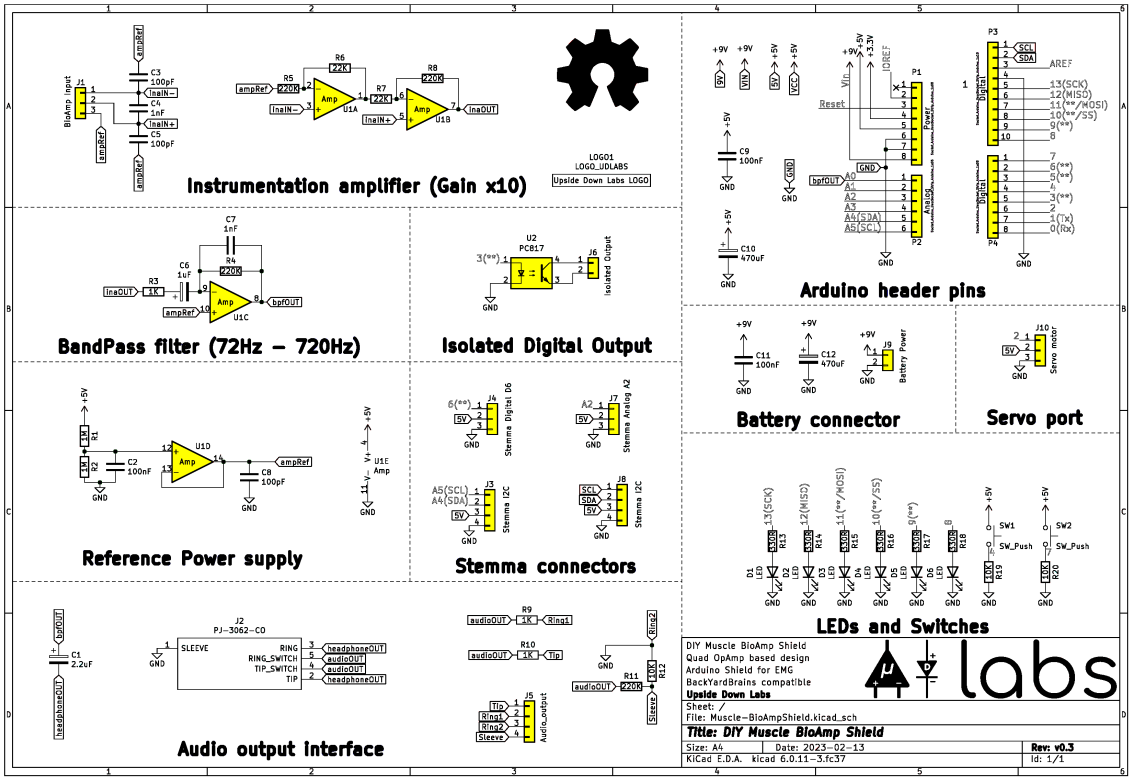


Fig. 5: *Schematic Diagram*

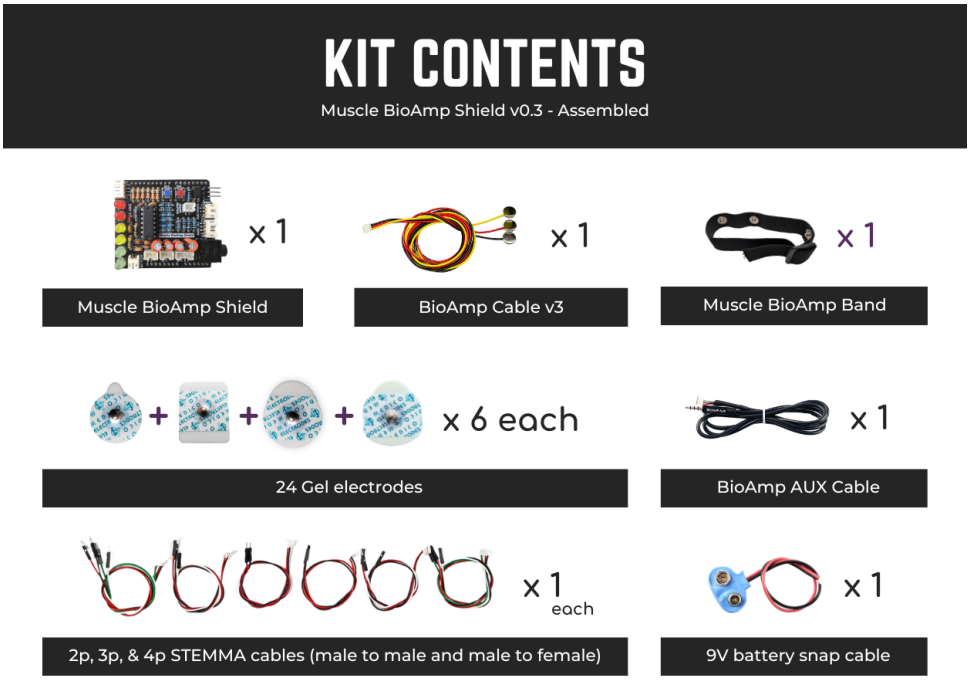
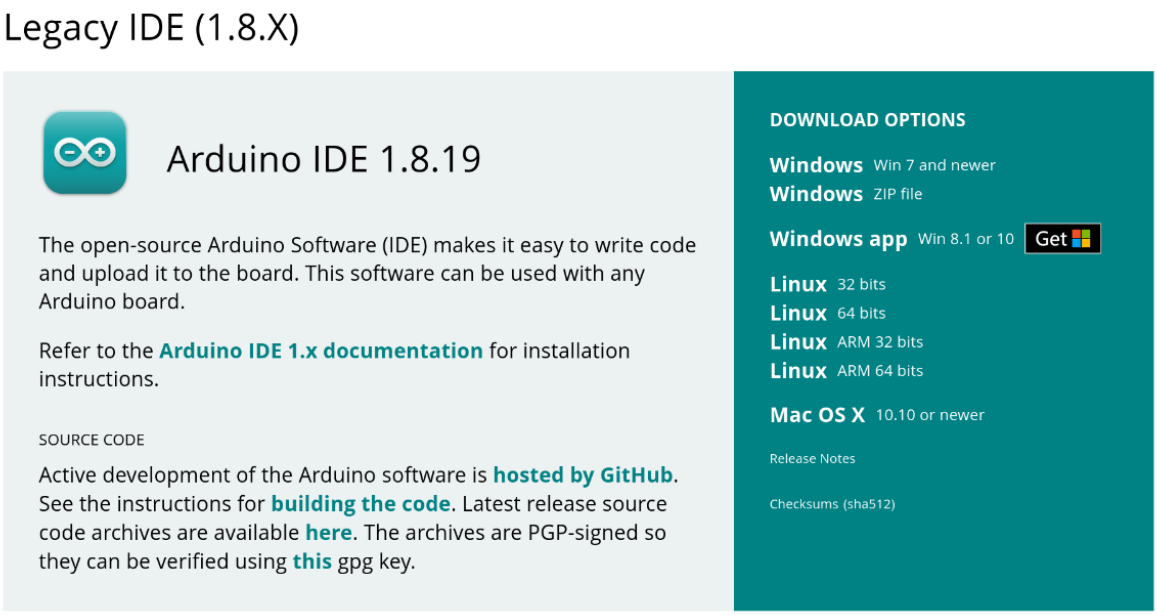


Fig. 6: *Assembled Muscle BioAmp Shield kit content*



Use Chords Web, our custom web interface designed to visualize biopotential signals data directly in your browser. *(Plug-and-play signal visualization directly in any modern web browser.)* Flash the Chords-Arduino-Firmware on your board to visualize signals on Chords Web.

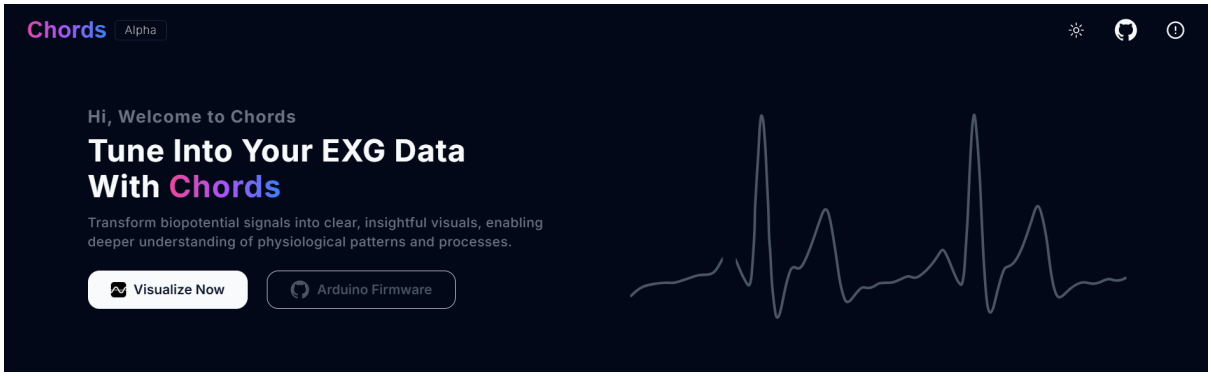


Fig. 7: Chords Web Landing Page

Note

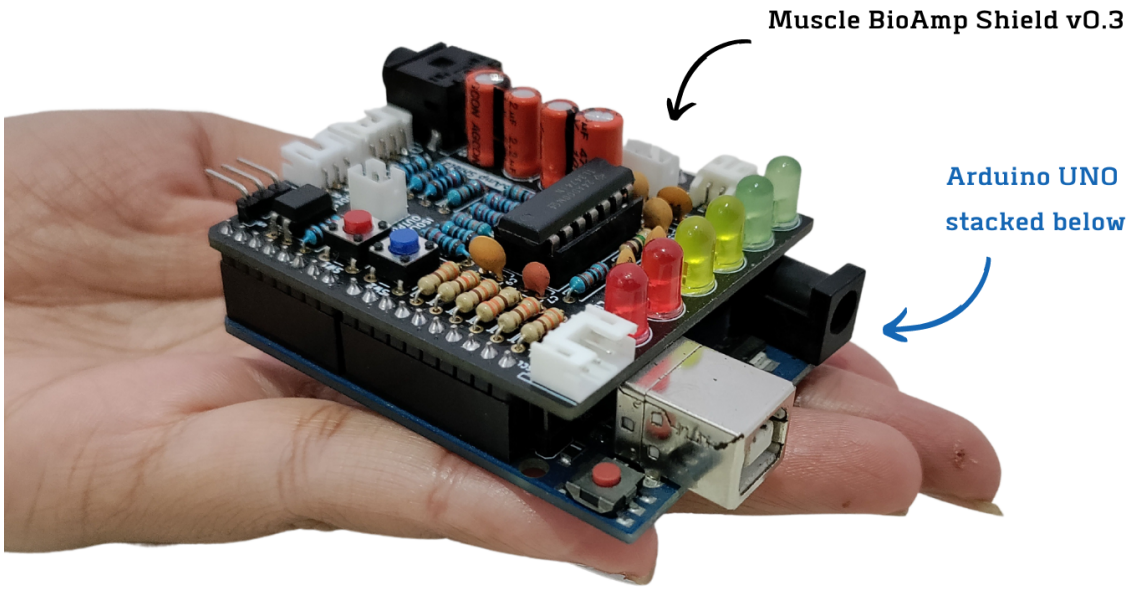
For detailed instructions and visualization tips, check the Chords Web Documentation.

https://youtu.be/IVIPnk9z75g

1.7 Using the Sensor

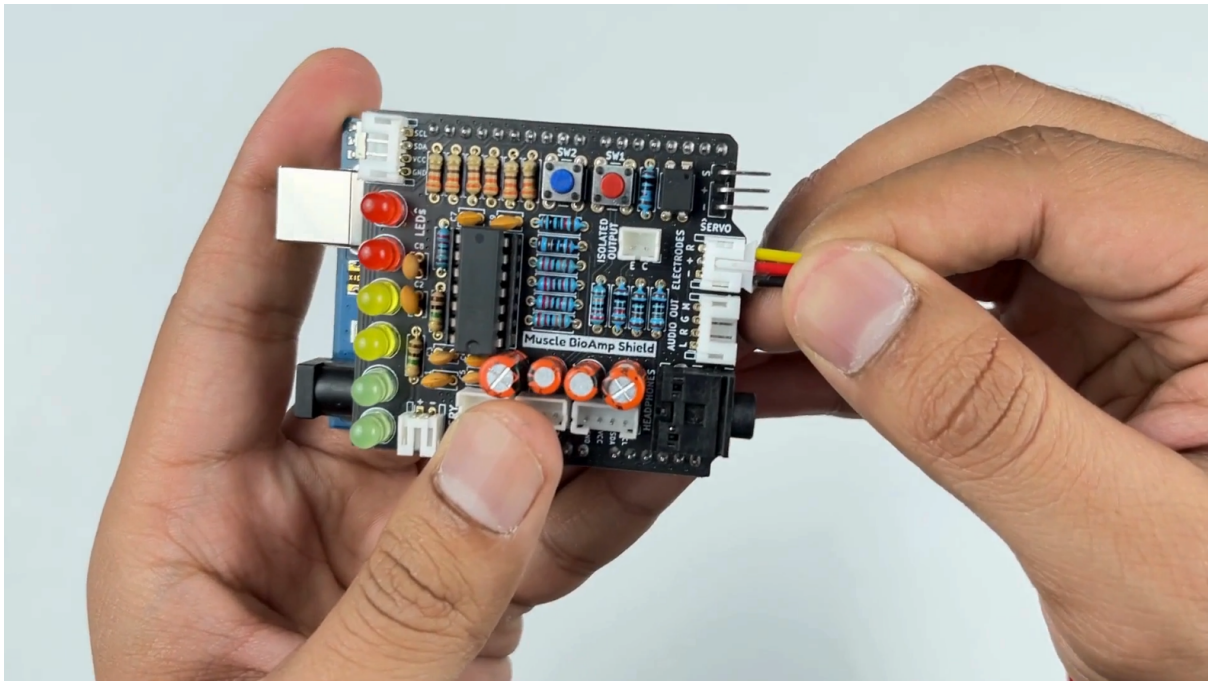
1.7.1 Step 1: Stack on Arduino Uno

Stack the Muscle BioAmp Shield on top of Arduino Uno properly.



1.7.2 Step 2: Connecting Electrode Cable

Connect the BioAmp Cable to Muscle BioAmp Shield as shown.



1.7.3 Step 3: Skin Preparation

Apply Nuprep Skin Preparation Gel on the skin surface where electrodes would be placed to remove dead skin cells and clean the skin from dirt. After rubbing the skin surface thoroughly, clean it with an alcohol wipe or a wet wipe.

For more information, please check out detailed step by step Skin Preparation Guide.

1.7.4 Step 4: Electrode Placements

We have 2 options to measure the EMG signals, either using the gel electrodes or using dry electrode based Muscle BioAmp Band. You can try both of them one by one.

Using gel electrodes

1. Connect the BioAmp cable to gel electrodes,
2. Peel the plastic backing from electrodes
3. Place the IN+ and IN- cables on the arm near the ulnar nerve & REF (reference) at the back of your hand as shown in the connection diagram.

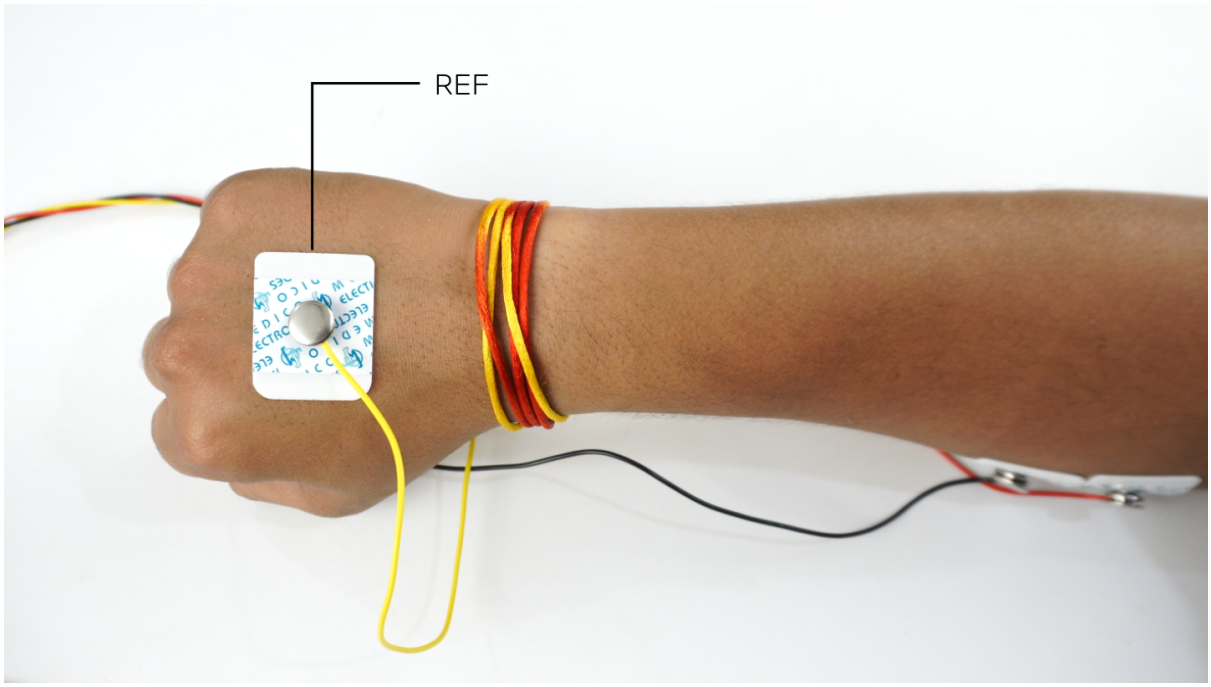


Fig. 8: *Electrode placement for REF cable*

Using Muscle BioAmp Band

1. Connect the BioAmp cable to Muscle BioAmp Band in a way such that IN+ and IN- are placed on the arm near the ulnar nerve & REF (reference) on the far side of the band.
2. Now put a small drop of electrode gel between the skin and metallic part of BioAmp cable to get the best results.

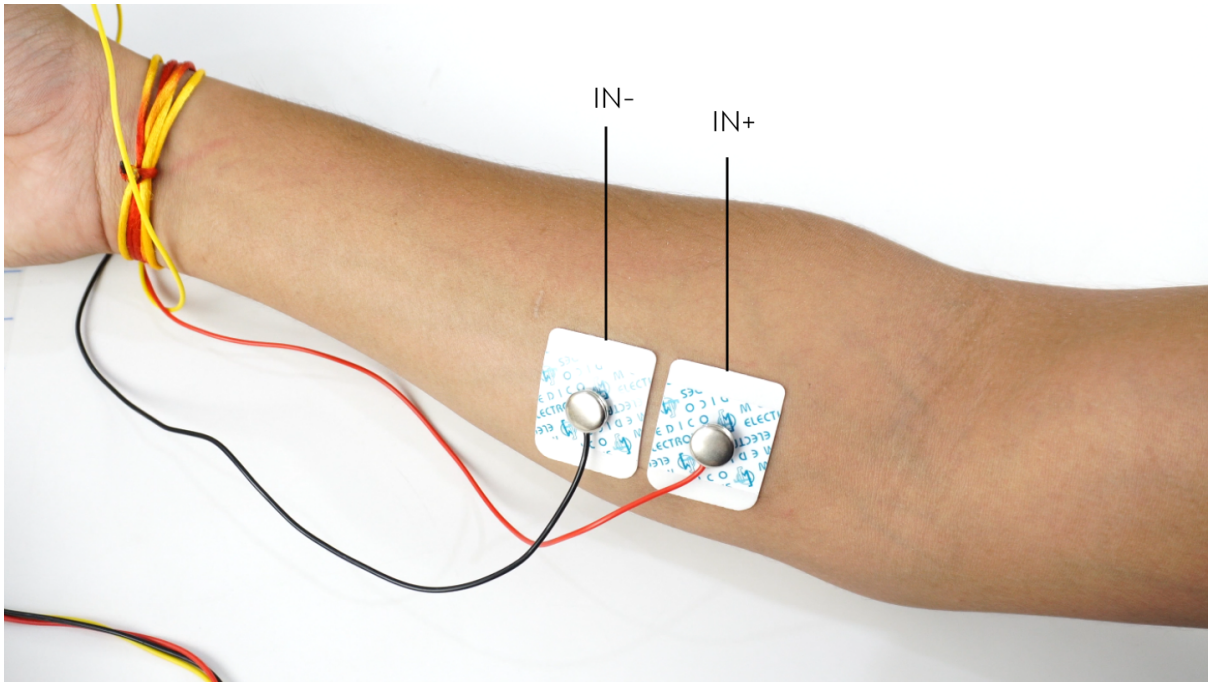


Fig. 9: *Electrode placement for IN+, IN- cables*

Tip

Visit the complete documentation on how to assemble and use the BioAmp Bands or follow the youtube video given below.

Tutorial on how to use the band:

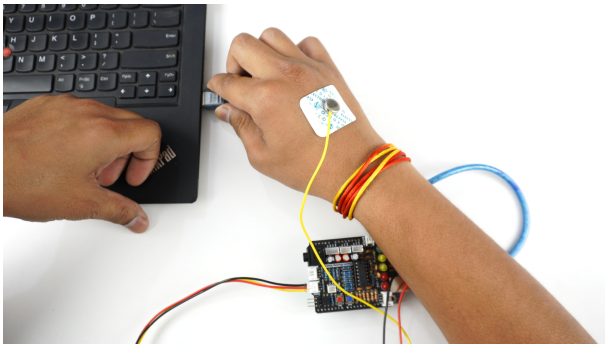
https://youtu.be/xYZdw0aesa0

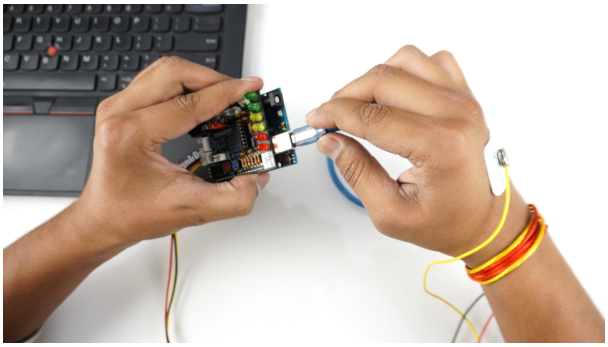
Note

In this demonstration we are recording EMG signals from the ulnar nerve, but you can record EMG from other areas as well (biceps, triceps, legs, jaw etc) as per your project requirements. Just make sure to place the IN+, IN- electrodes on the targeted muscle and REF on a bony part.

1.7.5 Step 5: Uploding the Code

- Connect your Arduino UNO to your laptop using the USB cable.





- Choose and copy one of the following Arduino sketches based on your requirements:

EMG Filter

EMG Envelope

To upload the code use Arduino IDE latest version (v2.3.8+):

- Go to `Tools` menu, navigate to `Board`, and select **Arduino UNO Development Board** (UNO R3/ R4/ Minima/ WiFi or any other board).
- In the same menu, select the `Port` to which your Arduino Uno is connected.
- **Tip:** To find the correct **COM Port**, disconnect your board and reopen the menu. The entry that disappears should be the right COM port.
- `Upload` the code.

Warning

For the best signal acquisition, ensure your laptop is not connected to a charger and maintain a distance of at least 5m from any AC appliances, read more about Tips for better signal acquisition.

1.7.6 Step 6: Visualise EMG signals on laptop

1. **Using Chords Web**

   For visualizing the EMG signals, use Chords Web for quick and hassle-free real-time bio-potential signals visualization right from your browser, without installing any software.

2. **Using Arduino legacy v1.8.19**

   To visualize the signal on Arduino v1.8.19 we will use inbuilt Serial Plotter:

- Navigate to (`Tools > Board`) and select your Development board or Arduino UNO Board (if not detected automatically).
- Open the Serial Plotter (`Tools > Serial Plotter` or `Ctrl+Shift+L`) to visualize the EMG signals.
- Once the Serial Plotter is open, ensure the baud rate is set to `115200`.
- Now flex your arm to visualize the muscle signals in real time on your laptop.

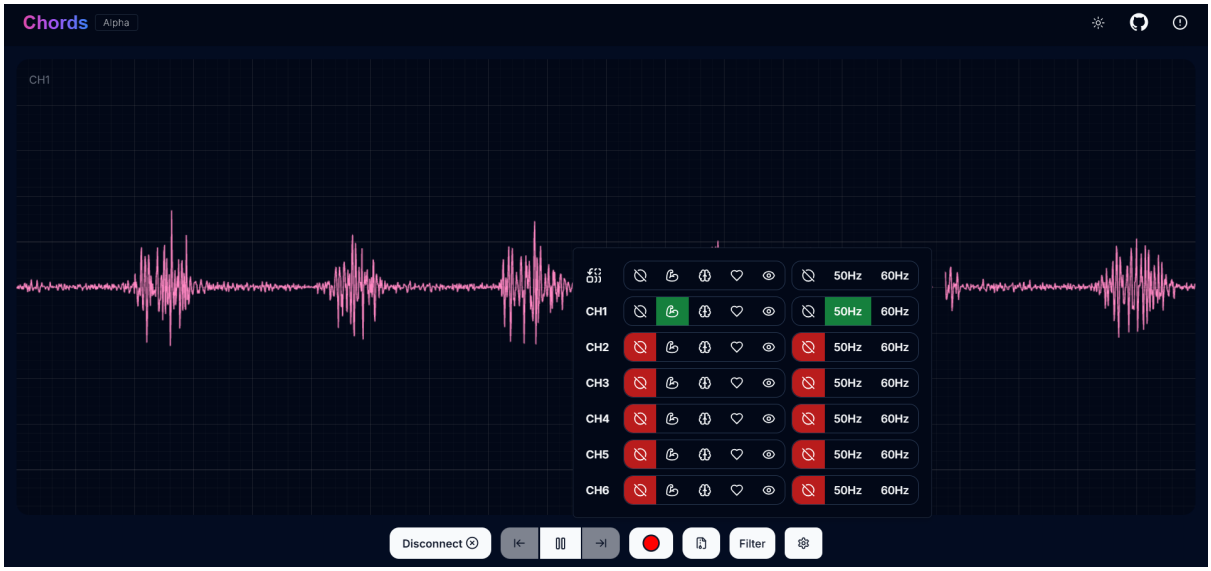


Fig. 10: Visualizing EMG signals on Chords Web

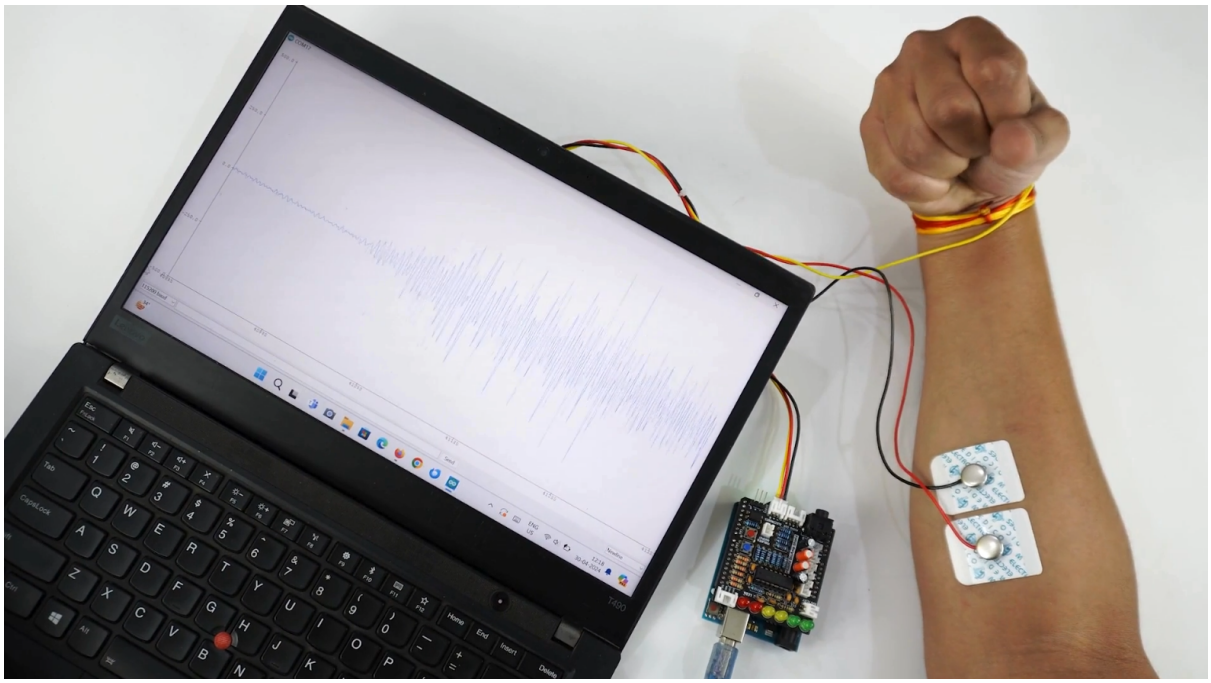


Fig. 11: *Visualise EMG signals on laptop*

1.7.7 Step 7: Visualise EMG signals on LEDs

Copy paste the Arduino Sketch given below in Arduino IDE:

LED Bar Graph

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm. You'll see the LED bar going up. More strength you apply, more the LED bar goes up.

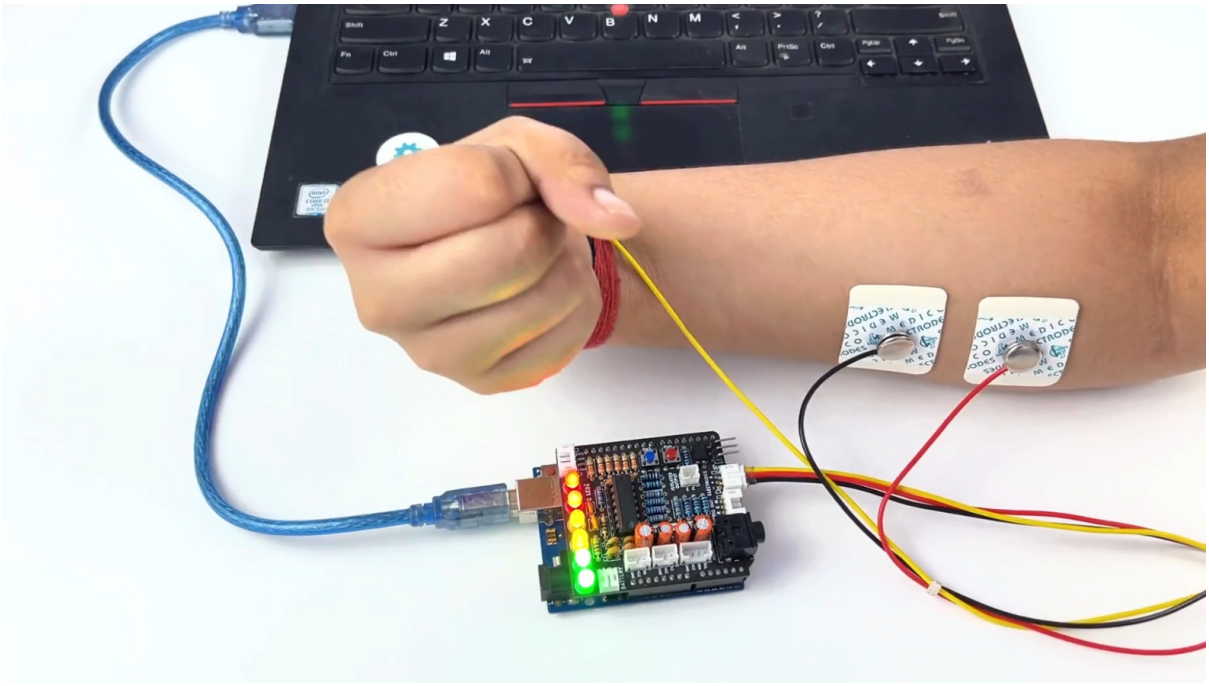


Fig. 12: *Visualise EMG signals on LEDs*

1.7.8 Step 8: Listen to your EMG signals

You can either listen to the muscle signals (EMG) on a speaker or wired earphones/headphones. Let's try both of them.

Listening EMG on speakers

1. Connect the BioAmp AUX cable on a bluetooth speaker that have 3.5mm jack support.
2. Switch on the speaker and turn the volume to maximum.
3. Flex and listen to your muscles.

Listening EMG on a wired earphones/headphones

1. Plug your wired earphones or headphones on the 3.5mm jack of BioAmp v1.5.
2. Plug it in your ears.
3. Flex and listen to your muscles.

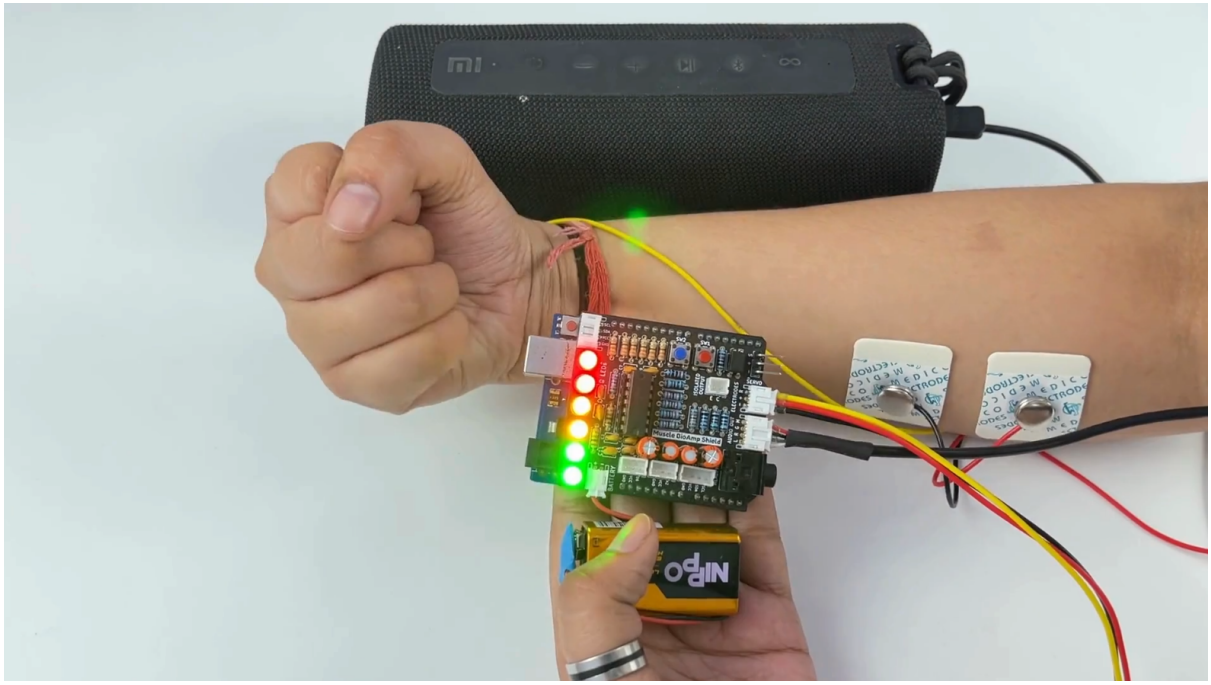


Fig. 13: Listening EMG on speakers

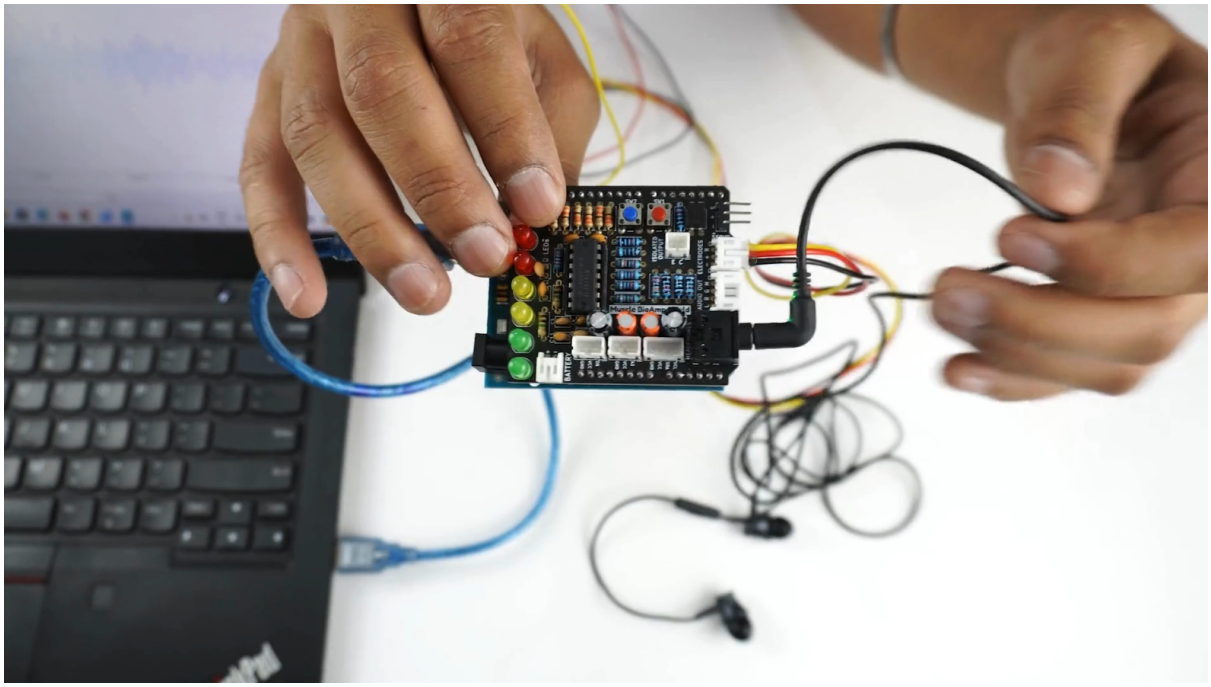


Fig. 14: *Listening EMG on a wired earphones/headphones*

1.7.9 Step 9: Controlling a servo motor

Connect the servo claw to Muscle BioAmp Shield.

Copy paste the Arduino Sketch given below in Arduino IDE:

Servo Controller

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm to control the servo claw in real time.

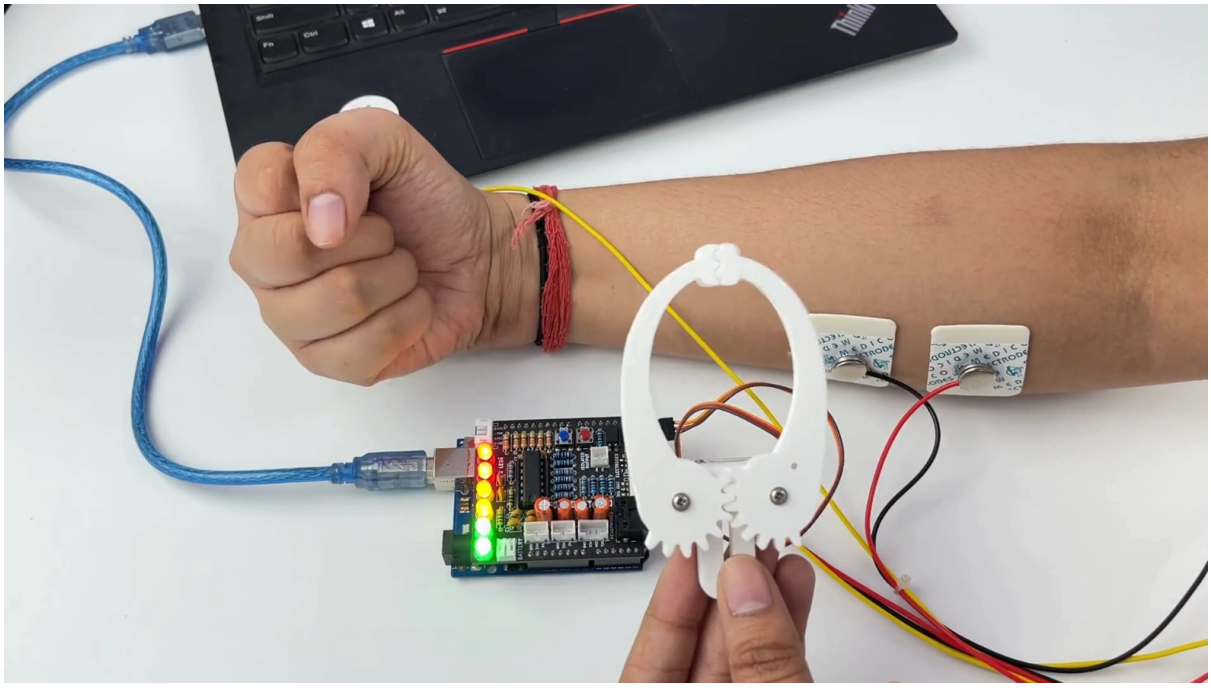
1.7.10 Step 10: Controlling a servo claw

Connect the servo claw to Muscle BioAmp Shield.

Copy paste the Arduino Sketch given below in Arduino IDE:

Claw Controller

Make sure you have selected the right board and COM port. Now upload the code, and flex your arm to control the servo claw in real time.



1.7.11 Step 11: Connecting 9V battery

Till now, the power for the EMG system was coming from the laptop via USB cable of Arduino Uno but there can be 2 ways in which you can make the system portable:

- **Using 9V battery:** Directly connect a 9V battery to Muscle BioAmp Shield using a 9V snap cable.
- **Using Power Bank:** Instead of connecting the USB cable of Arduino Uno to laptop, you can directly connect it to power bank.

Note

Do not use 9V battery while controlling a servo claw using Muscle BioAmp Shield. Instead connect the Arduino UNO to a power bank or directly to your laptop.

1.7.12 Step 12: Other functionalities you can explore

Using I2C ports

There are 2 I2C ports available on Muscle BioAmp Shield and you can connect hundreds of devices having I2C compatibility using the 4-pin JST PH 2.0 mm STEMMA cables provided.

Some of the examples are: `OLED screens`, `character displays`, `temperature sensors`, `accelerometers`, `gyroscopes`, `light sensors`, `BioAmp Hardware`, etc.

Using STEMMA Digital port

Connect Arduino Uno's D6 digital I/O pins using STEMMA digital connectors.

Using STEMMA Analog port

Connect Arduino Uno's A2 analog input pins using STEMMA analog connectors.

Using user buttons

Program the 2 user buttons according to your project requirements.